

HAT-P-32b

R-0034

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_180104 · created 2026-07-08T20:36:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458122.7616 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.054
Transit depth (Rp/Rs)^2	4.1e-06 +/- 0.0022 [%]
Orbital Inclination (inc)	86.86 +/- 1.73
Airmass coefficient 1 (a1)	0.62 +/- 0.41
Airmass coefficient 2 (a2)	0.4 +/- 0.28
Scatter in the residuals of the lightcurve fit is	63.38 %
Best Comparison Star	None
Optimal Aperture	1.7
Optimal Annulus	7.0
Transit Duration (day)	0.1137 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.054 vs 0.1489 (deviation 1.71σ)
Duration fit vs published	0.1137 d vs 0.1304 d (deviation 1.75σ)
Mid-time O–C	0.7 min
Depth SNR	0.0
Residual scatter	63.38 %

Light curve

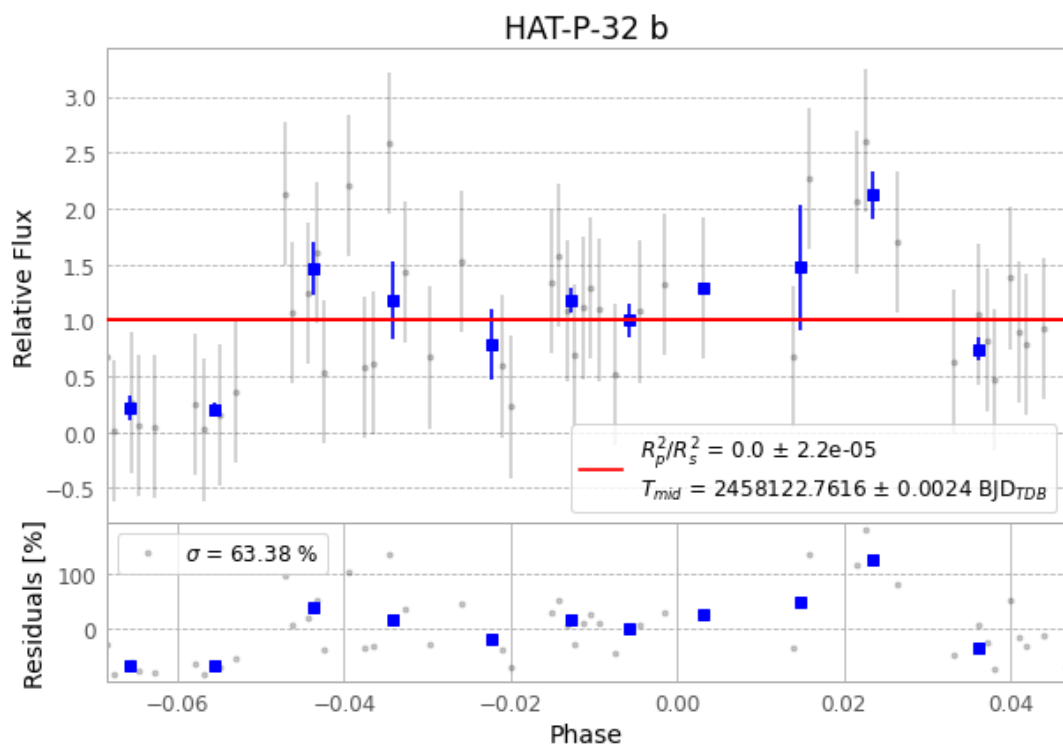


Figure 1 — FinalLightCurve_HAT-P-32 b_2018-01-03.png (SHA-256 f4beb8252e8bf517...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 140], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a5aa0be3955a56a1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

123 FITS frames (81 MB), HATP-32180104023913.FITS → HATP-32180104084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-180104.log (cb09ab8127ea...), FinalLightCurve_HAT-P-32 b_2018-01-03.png (f4beb8252e8b...), FinalLightCurve_HAT-P-32 b_2018-01-03.csv (8d6ab4ed8219...)

Compute resources

Wall time 208.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.